EARTHQUAKE ANALYSIS

PROJECT OVERVIEW

- This Tableau storytelling project presents an in-depth analysis of global earthquake activity, focusing on **magnitude, magnitude type, location, frequency, and seismic characteristics** across different years and geographic regions. The analysis explores how earthquake magnitudes vary over time and location, identifies years with the highest and lowest recorded magnitudes, and examines where earthquake activity is most and least concentrated.
- The project combines geographic, temporal, and statistical analysis to provide a structured view of earthquake behavior. It examines **maximum and minimum magnitude, magnitude type, earthquake count, depth, Dmin, gap, and RMS**, while allowing users to move from an overall view into a detailed location- and magnitude-level analysis. Interactive filters, parameters, clustering, and dashboard actions enable users to investigate specific years, magnitude types, magnitude ranges, and earthquake activity levels.
- The final storytelling experience is designed to help stakeholders understand historical earthquake patterns, identify areas of significant seismic activity, and use historical trends as a foundation for **situational awareness, risk assessment and future-oriented analysis**.

THE PROBLEM STATEMENT:

- The objective was to develop an interactive analytical solution that enables stakeholders to understand **when, where and how strongly earthquakes occurred** throughout the available historical dataset.
- The analysis needed to identify **all years and their recorded earthquake magnitudes, while specifically highlighting the Top 5 years based on maximum and minimum** magnitude. Stakeholders also required the ability to compare maximum and minimum magnitude observations geographically and understand how earthquake activity varied across different locations and magnitude types.

- The project further required analysis of important **seismic characteristics, including Depth, Dmin, RMS and Gap by magnitude type.** These measures provide additional context around the characteristics and distribution of recorded earthquake events. The analysis also needed to identify locations with the **highest and lowest numbers of earthquake occurrences,** determine the dominant magnitude types within those locations and drill down into the corresponding magnitude and seismic measurements.
- For future-oriented analysis, the project incorporates a **20-year forecast of aggregated magnitude trends. An additive trend-and-seasonality approach** was selected because it provides a straightforward way to model recurring patterns and underlying trends in historical time-series data while keeping the resulting forecast interpretable for dashboard users.

DATASET DETAILS

- The dataset was sourced from the **"Advanced Business Analytics" coursework** as part of the assigned mid-term project. The dataset contains **8,313 rows and 16 fields** covering earthquake events and their associated geographic, magnitude and seismic characteristics. The primary fields include:

  - Date
  - ID
  - Latitude
  - Longitude
  - Mag Type
  - Net
  - Place
  - Time
  - Time 1
  - Type
  - Depth
  - Dmin

- F4
- Gap
- Mag
- NST
- RMS

- During data preparation, **Time 1** was identified as a duplicate field and excluded. **NST and F4** contained null values and were also hidden from the analysis. Fields such as **Type and Net** were retained as descriptive attributes rather than analytical measures.
- The **MAG** field was treated as a continuous measure to support magnitude-based calculations and visual analysis.
- The dataset contains multiple magnitude types, including **MB, MJ, MS, MT, MW, MWB, MWC, MWP, MWR, MWW, and UK**, enabling comparisons of earthquake activity across different magnitude classifications.
- The supporting seismic measurements were interpreted according to the dataset structure:
  - **Depth:** earthquake depth, measured in meters
  - **Dmin:** minimum distance, measured in kilometers
  - **Gap:** seismic network gap, measured in kilometers
  - **RMS:** root mean square measurement, recorded in seconds

WHAT'S INSIDE

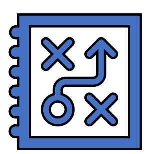
The project combines **historical, geographic, magnitude, seismic-measure and forecast analysis** into an interactive Tableau storytelling experience.

The dashboard collection includes:

- **Top 5 Maximum Magnitude Analysis** - identifies years with the highest recorded maximum magnitudes.
- **Top 5 Minimum Magnitude Analysis** - highlights years with the lowest recorded magnitude observations.
- **Yearly Magnitude Analysis** - compares maximum and minimum magnitude trends across the historical period.

- **Geographic Magnitude Clustering** - groups earthquake observations according to geographic location and aggregated magnitude.
- **Magnitude & Gantt Analysis** - connects earthquake location, magnitude type, magnitude, and year.
- **Magnitude Type & Seismic Analysis** - evaluates Depth, Dmin, Gap and RMS across magnitude types.
- **Forecast Analysis** - extends the historical aggregated magnitude trend into a 20-year forecast.
- **Earthquake Count by Place** - ranks locations according to the number of recorded earthquake events.
- **Place & Magnitude-Type Deep Dive** - identifies the dominant magnitude types within individual locations.
- **Maximum & Minimum Magnitude Comparison** - compares magnitude ranges across locations and magnitude types.
- **Interactive Highest/Lowest/All Places Analysis** - allows users to switch between locations with the highest earthquake activity, lowest activity, or all available locations.
- **Detailed Crosstab Analysis** - Provides earthquake counts, maximum/minimum magnitude, and magnitude-type breakdowns for each location.

The storytelling flow moves from **"When and how strong?" → "Where?" → "What magnitude type?" → "What are the associated seismic characteristics?" → "What does the future trend indicate?"**



◆ DAX FORMULAS & NEW MEASURES

❖ **Magnitude Cluster :** A calculated field named **"Mag Cluster"** was created to categorize earthquakes according to their magnitude range. The classification progresses through **Clusters 1, 2, 3 and 4**, representing increasing magnitude levels, with the highest magnitude observations assigned to Cluster 4. This clustering provides an additional analytical layer for the geographic and magnitude analysis.

❖ **Earthquake Count Selection Parameter :** An **"Earthquake Count Selection"** parameter was created with three options:

- **Highest**
- **Lowest**
- **All Places**
- This allows users to switch **between locations with the highest earthquake activity, locations with the lowest earthquake activity, or the complete set of locations.**

❖ **Earthquake Count by Place :** A calculated field was created to determine the number of earthquake records associated with each location using the earthquake **ID**.
This measure forms the foundation for identifying locations with the highest and lowest earthquake activity.

❖ **Place Selection Filter :** A calculated field was connected to the Earthquake Count Selection parameter to dynamically control the locations displayed in the dashboard.
When **All Places** is selected, all locations remain visible. When **Highest** is selected, the analysis focuses on locations with the maximum earthquake count. When **Lowest** is selected, the analysis identifies locations with the minimum earthquake count.

❖ **Earthquake Count by Magnitude Type :** Earthquake counts were analyzed by magnitude type and sorted in descending order to highlight the magnitude classifications contributing the most earthquake observations within each location.

◆ DATA MODELING

➢ The earthquake dataset was first loaded into Tableau and the field data types were reviewed and adjusted according to their analytical purpose. Duplicate and unusable fields were removed from the analysis, including **Time1**, which duplicated the time information, and **NST** and **F4**, which contained null values. Fields such as **Updated** were excluded because they were not required as analytical measures for the project.

➢ The **MAG** field was converted from a dimension into a **continuous measure**, allowing it to be used effectively for maximum, minimum, sum, and other magnitude-based calculations across the visualizations.

➢ For the majority of the visualizations, **Year(Date)** was maintained as a **discrete field** so that individual years could be clearly compared and selected. For the forecasting analysis, Year(Date) was treated as a **continuous time field** to maintain a chronological time series and support the 20-year forecast.

VISUALS UTILIZED TO ANALYZE THE DATA

- The project combines multiple Tableau visualization techniques to analyze earthquake activity from different perspectives.

- **Top 5 Magnitude Analysis :** Packed bubble charts were created to identify the **Top 5 years based on maximum magnitude** and the **Top 5 years based on minimum magnitude**. Year-based filtering allows users to focus on the most significant magnitude observations within the historical dataset.

- **Yearly Magnitude Distribution :** A **Box-and-Whisker chart** was used to examine the distribution of earthquake magnitude by year and highlight the variation in maximum magnitude across the historical period.

- **Geographic Magnitude Cluster Analysis :** A geographic cluster analysis was developed to visualize the distribution of maximum earthquake magnitude across locations. The magnitude clusters provide an additional visual interpretation of earthquake intensity across geographic areas.

- **Magnitude and Place Timeline :** A **Gantt chart** was used to analyze the relationship between **place, magnitude, magnitude type, and year**, providing a temporal view of earthquake events.

- **Seismic Measure Analysis :** Dual-axis visualizations were used to analyze seismic characteristics by magnitude type:
  - **Depth and Dmin:** stacked bar and line combination
  - **Gap and RMS:** line and circle combination .
  - These visuals allow users to compare different seismic measures while maintaining a clear distinction between the underlying metrics.

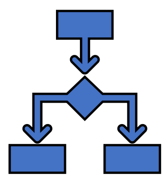
- **Forecast Analysis :** A line chart was used to present the historical aggregated magnitude trend and extend the analysis into a **20-year forecast**, providing an estimate of potential future magnitude patterns based on historical behavior.

- **Earthquake Count by Place :** A horizontal bar chart was created to rank locations according to their **number of earthquake records**, sorted from highest to lowest activity.

- **Earthquake Count by Magnitude Type :** Magnitude types were further analyzed within each location to determine which classifications contributed the most earthquake observations.

- **Maximum and Minimum Magnitude by Place :** A dual-axis line and circle visualization was developed to compare **maximum and minimum magnitude across locations**, allowing users to understand both the magnitude range and variation between places.

- **Detailed Earthquake Crosstab :** A detailed crosstab was created to show:
  - Earthquake count by magnitude type
  - Maximum magnitude
  - Minimum magnitude
  - Total earthquake count
  - Magnitude types within each location .
  - The table is sorted by earthquake count in descending order so that locations and magnitude types with greater activity appear first.



STEPS I FOLLOWED TO BUILD THE DASHBOARD

1. **Max Magnitude**: A dashboard named "Max Magnitude" was created to analyze earthquake magnitude across all available years. The dashboard highlights the **Top 5 years based on maximum magnitude and provides an interactive view of the strongest recorded earthquake years**. Dashboard actions were configured so that selections from the primary visualization could filter the supporting worksheets and maintain interaction across the analysis.

2. **Magnitude Cluster & Gantt Dashboard** : A dedicated Magnitude Cluster and Gantt dashboard was created to **combine geographic magnitude clustering with temporal earthquake analysis**. Interactive filters were added for:

- Year
- Magnitude Type
- Magnitude
- Magnitude Cluster

- Dashboard actions allow users to select information from one visualization and dynamically filter the related views.

3**. Magnitude Type & Year Analysis Dashboard** : A separate dashboard was developed to examine seismic characteristics by magnitude type and year. The dashboard includes:

- Depth and Dmin analysis using a dual-axis stacked bar and line visualization
- RMS and Gap analysis using a dual-axis circle and line visualization
- Magnitude-type comparisons
- Historical trend analysis
- This dashboard provides additional context around the characteristics associated with different earthquake magnitude classifications.

4. **Deep Dive Earthquake Analysis Dashboard** : The final dashboard provides a detailed location-level analysis of earthquake activity. It incorporates:

- Earthquake count by place
- Maximum and minimum magnitude
- Magnitude type
- Maximum/minimum magnitude by location
- Earthquake count by magnitude type
- Magnitude range by magnitude type
- Geographic cluster analysis
- Interactive dashboard actions were configured between source and target worksheets so users can select a location, magnitude type, or cluster and investigate the corresponding details.
- **The Earthquake Count Selection parameter allows the user to switch between:**

  Highest → Lowest → All Places

- This makes it possible to compare high-activity and low-activity locations while examining their associated magnitude types and seismic characteristics. Additional filters were provided for Magnitude Cluster, Magnitude Type, Maximum Magnitude, Minimum Magnitude, Earthquake Count and location selection, creating a multi-level drill-down experience.

THE STORYTELLING

- The overall Tableau story is designed to progress through four analytical levels:

Historical Magnitude → Geographic Distribution → Magnitude Type & Seismic Characteristics → Location-Level Deep Dive.

- This allows the viewer to start with when significant earthquakes occurred, understand where they were concentrated, examine how **magnitude types and seismic characteristics differed, and finally drill down into specific locations** to understand their earthquake frequency and magnitude behavior.
- The result is not just a collection of charts, but an interactive analytical **narrative that connects time, magnitude, geography, frequency, and seismic characteristics** into a single exploration experience.

WHY THIS PROJECT MATTERS

- Earthquake datasets contain **multiple dimensions- time, location, magnitude, magnitude type, frequency and seismic measurements**- that can be difficult to interpret when viewed independently. This project brings those dimensions together into a single interactive analytical experience.
- The analysis helps stakeholders quickly identify **historically significant earthquake years, high- and low-activity locations, dominant magnitude types, magnitude ranges and differences in seismic characteristics.** Instead of looking only at the magnitude of an earthquake, the dashboard provides additional context through **Depth, Dmin, Gap, RMS, geographic clustering and event frequency.**
- The forecasting component **further extends the analysis by providing a view of the potential future direction of the aggregated magnitude trend.** While the forecast should be interpreted as a data-driven projection rather than a prediction of specific future earthquakes, it provides a useful way to explore historical patterns and potential trends.

KEY INSIGHTS

- The analysis identified earthquake magnitudes ranging from approximately **6.0 to 9.6**, demonstrating substantial variation in the intensity of recorded earthquake events.

Maximum Magnitude by Year

The Top 5 years with the highest recorded maximum magnitude were:

Year	Maximum Magnitude
1960	**9.6**
1964	**9.2**
2004	**9.1**
1952	**9.0**
2011	**9.0**

Among these, **1960 recorded the highest maximum magnitude at 9.6**.

- The minimum-magnitude analysis identified **1964–1968** among the Top 5 years, with the minimum magnitude remaining around **6.0**. The yearly minimum-magnitude trend also showed that the minimum magnitude remained at approximately 6.0 through the later portion of the dataset.
- The maximum-magnitude trend showed values beginning around **7.7 in 1900** and reaching approximately **6.9 by 2014** in the displayed annual series.

Geographic Clustering

- A four-cluster geographic analysis was developed using **Sum of MAG** at the **Latitude and Longitude** level. The clustering used normalized variables and analyzed **8,197 points**.

The clustering produced four groups:

- **Cluster 1:** 8,116 points, Sum of MAG = 6.5563
- **Cluster 2:** 61 points, Sum of MAG = 13.775

- **Cluster 3:** 18 points, Sum of MAG = 22.894
- **Cluster 4:** 2 points, Sum of MAG = 51.75

➢ The results show that the majority of **observations belong to Cluster 1, while the smaller higher-level clusters contain comparatively fewer observations** but substantially higher aggregated magnitude values.

❖ **Magnitude Type Analysis**

➢ The magnitude types **MB, MS and MW** accounted for a large portion of the earthquake locations represented in the dataset.

➢ The seismic-measure analysis also highlighted differences between magnitude types:

- **MS** recorded the highest aggregated Depth at approximately **286,024**.
- **MWR** recorded the lowest aggregated Depth at approximately **40**.
- **MWP** had the lowest aggregated Dmin at approximately **16.2**.
- **MW** had the highest aggregated Dmin at approximately **252.4**.
- **MWR** recorded the lowest aggregated Gap at approximately **31**.
- **MWC** recorded the highest aggregated Gap at approximately **14,658**.
- **MWR** recorded the lowest aggregated RMS at approximately **0.9**.
- **MWC** recorded the highest aggregated RMS at approximately **514.0**.

These comparisons provide additional context beyond magnitude alone by showing how the associated seismic measurements vary across magnitude classifications.

DEEP-DIVE LOCATION INSIGHTS

➢ A major part of the analysis focused on identifying locations with the highest and lowest earthquake activity.

➢ The analysis identified **Vanuatu as the location with the highest earthquake count, with 349 recorded earthquake events**. The earthquakes associated with Vanuatu ranged from a **minimum magnitude of 6.0 to a maximum magnitude of 7.9**, across magnitude types including **MB, MS, MWB, MWC, MWW and UK**.

➢ When the analysis was drilled down to magnitude type, **MS was the dominant magnitude type in Vanuatu**, accounting for **166 earthquake records**, with a maximum magnitude of **7.5** and a minimum magnitude of **6.0**.

- The analysis also identified multiple locations with only **one recorded earthquake**, representing the lowest observed earthquake count in the dataset.
- This led to the development of an interactive **Earthquake Count Selection** parameter that allows users to switch between: **Highest → Lowest → All Places.**
- This enables the dashboard to move from a high-level geographic comparison into a targeted investigation of individual locations.

CHALLENGES & HOW THEY WERE ADDRESSED

- Throughout the development of the Tableau storytelling project, several analytical and visualization challenges were encountered. Each challenge required refining the calculations, dashboard structure, and user interactions to ensure that the final analysis accurately represented the earthquake data.

1. Identifying Locations with the Highest and Lowest Earthquake Counts

- One of the initial challenges was determining **which locations experienced the highest and lowest number of recorded earthquakes**. Since the dataset contained numerous place names, simply displaying the locations was not sufficient.
- The analysis identified **Vanuatu as the location with the highest earthquake count, with 349 recorded events**. The earthquakes recorded there ranged from a minimum magnitude of **6.0** to a maximum magnitude of **7.9**, across magnitude types including **MB, MS, MWB, MWC, MWW and UK**.
- At the opposite end, multiple locations had the lowest recorded earthquake count of **one event**. To make this analysis interactive, an **Earthquake Count Selection** parameter was introduced with options for **Highest, Lowest and All Places**. This allows users to move between the most active locations, least active locations, and the complete set of locations without requiring separate visualizations.

2. Drilling Down from Place to Magnitude Type

- The next challenge was to determine **which magnitude types contributed to earthquake activity within each individual place**.
- Initially, adding Mag Type directly to the visualization resulted in similar or misleading counts across magnitude types because the calculation was not being evaluated at the required **Place + Mag Type** level.
- To address this, a calculated field named **Earthquake Count by Mag Type** was created to calculate the number of earthquake IDs associated with each magnitude type within each location.

- This enabled the analysis to move from:

Place → Total Earthquake Count

to:

Place → Magnitude Type → Earthquake Count

3. Showing Both Magnitude-Type Counts and Overall Place Counts

- Another challenge occurred when the requirement was expanded to show both:
  - The number of earthquakes within each magnitude type
  - The total number of earthquakes within the corresponding place
- Adding both calculations directly to the visualization caused overlapping values and reduced readability. To resolve this, a separate **Earthquake Count by Place** calculation was added and the Tableau **Analysis → Totals → Add All Subtotals** functionality was applied.
- This allowed the visualization to show the earthquake count for each magnitude type while also presenting the **overall earthquake count as a subtotal for each place**, making the hierarchy much easier to interpret.

4. Comparing Maximum and Minimum Magnitude by Magnitude Type

- The next analytical challenge was determining: **Which magnitude type had the highest and lowest magnitude within each place?.** The requirement was not simply to display the overall maximum and minimum magnitude. The analysis needed to identify the **maximum and minimum magnitude within each individual magnitude type and place**.
- To address this, the relevant measure values were separated into **MAX(MAG)** and **MIN(MAG)** and incorporated into the detailed place-level visualization.
- This created a clearer comparison such as:

Place → Magnitude Type → Earthquake Count → Maximum MAG → Minimum MAG

- The overall place-level earthquake count was also retained, allowing the viewer to understand both **frequency and magnitude range** within the same analytical view.
- For example, within **Vanuatu**, the **MS magnitude type** was the most frequently recorded, with **166 earthquake events**, ranging from a minimum magnitude of **6.0** to a maximum magnitude of **7.5**.

5. Adding Additional Seismic Measures

➢ The analysis was then extended beyond earthquake count and magnitude to examine additional seismic characteristics:

- **Depth**
- **Dmin**
- **Gap**
- **RMS**

➢ These measures were incorporated into the detailed analysis by magnitude type and place to provide additional context around the characteristics of the recorded earthquake events. However, this introduced another visualization challenge.

6. Handling NULL Values Without Leaving Empty Measures

➢ Several earthquake records contained missing values for Depth, Dmin, Gap, or RMS. Displaying these fields directly resulted in blank measure values remaining within the visualization, creating unnecessary gaps and reducing readability.

➢ Instead of replacing missing values with zero which could incorrectly imply that the measurement was actually zero the data structure was redesigned using a **pivot-based Measure / Value approach**.

➢ The relevant measures were pivoted into:

- **Measure**
- **Value**

➢ NULL values were then excluded from the visual analysis. This allowed the visualization to dynamically display only the measures that contained actual values for a particular **Place + Magnitude Type** combination.

➢ For example, if Dmin was unavailable for a specific magnitude type, the visual would move directly from the preceding available measure to the next available measure rather than displaying an empty Dmin row. This resulted in a cleaner and more meaningful deep-dive visualization.

7. Extending the Analysis into Forecasting

- The final challenge involved extending the historical magnitude analysis into a future-oriented view. The actual aggregated MAG series increased from approximately **149 in 1900** to **917 by 2013**. The forecast begins in **2014**, with an estimated aggregated magnitude of approximately **1,058**, and extends through **2033**, reaching approximately **1,262**. This forecast provides an analytical extension of the historical trend and helps the viewer explore how the aggregated magnitude measure could evolve based on the historical time series.

◆ OVERALL CHALLENGE-RESOLUTION JOURNEY

- The challenges progressively shaped the final dashboard architecture:

Place Count → Magnitude Type → Magnitude Range → Seismic Measures → NULL Handling → Forecasting.

- Rather than presenting only a high-level earthquake count, the final solution enables users to drill down from **where earthquakes occurred most or least frequently** to **which magnitude types dominated those locations**, their **maximum and minimum magnitudes**, and the associated **Depth, Dmin, Gap, and RMS measurements**.
- This iterative process also strengthened the dashboard's interactivity through parameters, calculated fields, subtotals, dashboard actions and dynamic filtering, resulting in a more structured and user-focused earthquake storytelling experience.

❖ **STRATEGIC RECOMMENDATIONS**

❖ **1. Prioritize high-activity locations for deeper monitoring**

- Locations with substantially higher earthquake counts should receive greater analytical and monitoring attention. **Vanuatu**, with 349 recorded events in this dataset, represents a clear example of a high-activity location that warrants deeper examination.

❖ **2. Analyze dominant magnitude types separately**

- Because different magnitude types exhibit different distributions and seismic characteristics, stakeholders should avoid treating all earthquake observations as identical. The dominant magnitude types within each location should be analyzed separately when assessing historical activity.

- **3. Use magnitude range rather than maximum magnitude alone**
    - A location's maximum magnitude provides only one part of the story. Comparing **minimum and maximum magnitude together** gives stakeholders a better understanding of the observed magnitude range and helps distinguish locations with consistently higher activity from those with isolated extreme events.

- **4. Combine earthquake frequency with intensity**
    - A location with a large number of earthquakes is not necessarily the location with the strongest earthquakes. Therefore, **earthquake count and magnitude should be evaluated together**. For example, the dashboard can identify: **Where earthquakes occurred most frequently?** and separately: **Where the strongest earthquakes occurred?**
    - Combining these perspectives provides a more balanced assessment of earthquake activity.

- **5. Incorporate seismic characteristics into location assessment**
    - Depth, Dmin, Gap, and RMS should be considered alongside magnitude and frequency. The differences observed across magnitude types suggest that these supporting measures can provide additional context when evaluating earthquake events.

- **6. Use geographic clustering to prioritize analysis**
    - The clustering results can be used as an exploratory method for identifying areas with different levels of aggregated magnitude activity. Locations belonging to smaller, higher-magnitude clusters can be investigated further rather than relying solely on overall earthquake counts.

- **7. Use the forecast for scenario planning—not exact earthquake prediction**
    - The 20-year forecast can support **long-term analytical planning and scenario development**, but it should not be interpreted as predicting the exact timing or location of future earthquakes. Earthquake occurrence is highly complex, so the forecast should remain a trend-oriented analytical component of the dashboard.

- **8. Maintain the interactive drill-down approach**
    - The **Highest / Lowest / All Places** parameter is particularly useful because it allows stakeholders to move between broad and focused analysis without creating separate dashboards for every scenario. This approach should be retained when extending the project with additional earthquake datasets.

- **9. Use the dashboard as an analytical starting point**
    - The project should be viewed as a **decision-support and exploratory analytics tool** rather than a standalone disaster-prediction system. Its greatest value is in helping users quickly discover patterns, compare locations, identify unusual observations, and determine where additional investigation may be required.